A Sparsity Prior for Attention Pooling: Refuted, Diagnosed, and Corrected

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Order-Dependence in a Stick-Breaking Variational Bottleneck | September 2026

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Abstract

We test whether replacing fixed mean-pooling with a variational mixture over tokens — weights drawn from a reparameterizable stick-breaking process, content from a Gaussian posterior, both regularized by a single KL term — improves a classifier's generalization to sequences longer than any seen in training. The direct test refutes the hypothesis: at eight times the training length, the mixture is markedly less accurate than a mean-pooling baseline (0.567 vs. 0.882). Splitting that number by where the informative token happens to fall reveals why: the mixture is dramatically *more* accurate than the baseline when the signal is early in the sequence and far less accurate when it is late, because stick-breaking allocates mass by position as well as content, and later positions receive vanishingly little regardless of what they contain. We measure this distortion directly (a 74× ratio between the mass given to the first and last quarter of a length-128 sequence, against a uniform baseline of 1×), confirm it is the cause with a causal intervention that recovers the accuracy it costs, and then show that a hyperparameter believed to control sparsity in fact controls the opposite quantity: raising it removes most of the distortion without any architectural change, reaching parity with the baseline at a value the initial experiment did not use. We report the refuted hypothesis, the correct diagnosis, and the hyperparameter correction in the order they were found.

1 Motivation

Fixed pooling operators — mean, max, a designated summary token — give every input the same fixed-size treatment regardless of how much of it is actually informative. A variational alternative can in principle allocate its capacity adaptively: commit weight to few tokens when few are relevant, spread it out when many are. The natural question for such a mechanism is whether that adaptive commitment transfers to inputs longer than anything seen during training, where a fixed-capacity pooling operator has no mechanism to compress what it has not been sized for.

2 Mechanism

Given per-token hidden states $H \in \mathbb{R}^{B\times T\times D}$ from a bidirectional Transformer encoder, each token emits stick-breaking parameters (a, b) and Gaussian posterior parameters $(\mu, \log \sigma^2)$. Reparameterizable stick proportions v_k are drawn from a Kumaraswamy distribution — the relaxation that makes stick-breaking differentiable end-to-end, avoiding a high-variance score-function estimator — and combined into mixture weights $\pi_k = v_k \prod_{j<k}(1 - v_j)$, with prior $\mathrm{Beta}(1, \alpha_0)$. Each token's content is sampled $z_t = \mu_t + \sigma_t \epsilon$. The pooled representation is $\sum_t \pi_t z_t$, and a single KL term regularizes both the mixture weights toward their prior and the content posteriors toward $\mathcal{N}(0, I)$, exactly as in a standard variational information bottleneck objective.

Task. A synthetic sequence-classification problem in which exactly one token determines the label and the remainder is uninformative noise. Models are trained at sequence length 16 and evaluated at $\{16, 32, 64, 128\}$ — the signal is held constant while the amount of noise diluting it grows by 8×. The ground-truth informative position is known by construction, which is what makes the diagnosis in Section 4 possible. Baseline: an identical encoder with uniform masked mean-pooling in place of the bottleneck, so the pooling operator is the only variable.

3 Result: The Direct Test

Three seeds, sequence length 16 training, $\alpha_0 = 5$. Held-out accuracy, mean $\pm$ std:

Test length	Baseline	Stick-breaking bottleneck
16 (train)	1.000 ± 0.000	1.000 ± 0.000
32	1.000 ± 0.000	0.995 ± 0.004
64	1.000 ± 0.000	0.822 ± 0.089
128	0.882 ± 0.159	0.567 ± 0.111

Table 1: Both models solve the training length perfectly; the bottleneck degrades faster out of distribution. Read alone, this refutes the hypothesis.

Taken alone, Table 1 says the variational mixture hurts length generalization. That conclusion turns out to be incomplete.

4 Diagnosis: Position, Not Content, Predicts Failure

Measuring the mixture weights against the known informative-token position shows the mechanism is working exactly as intended *on average*: at training length the bottleneck places 83% of its mass on the single correct token and identifies it in 100% of examples, with concentration (mass relative to chance) rising from 13.3× at length 16 to 25.7× at length 128.

What fails is where that mass is available to go. Averaging mixture weight by absolute position, independent of content: at length 128 the first quarter of the sequence holds 99% of all mixture mass and the last quarter holds 0.0% (Figure 1, right) — a 74× distortion against the 1× a content-blind pooling operator would show. Stick breaking is order-dependent by construction (π_k can only be large if v_j for all $j < k$ were small), so a token's position, not only its content, determines how much mass it can ever receive. Splitting accuracy by where the informative token happens to fall confirms this is the whole story: at length 128 the bottleneck scores 1.000 against the baseline's 0.738 when the signal is in the first quarter, and 0.362 against 0.693 when it is in the last — a large win and a large loss that the aggregate number in Table 1 averages into a single, more pessimistic figure.

5 Causal Confirmation

Correlation between position-dependence and the accuracy failure is not yet causation. As a direct intervention, stick order is randomized independently per example at each forward pass, restoring exchangeability in expectation while changing nothing else about the model or its training. This collapses the positional ratio from 74× to 0.7× — effectively uniform — and recovers 15.4 accuracy points at length 128 (0.567 → 0.721) and 15.6 points at length 64 (0.822 → 0.978), with no architectural change beyond the order of one internal computation.

Randomizing an internal ordering that should be irrelevant to the model's stated computation recovers most of the length-generalization gap, which is the causal signature of the diagnosis in Section 4.

Shuffling does not fully close the gap to the mean-pooling baseline at length 128 (0.721 vs. 0.882):

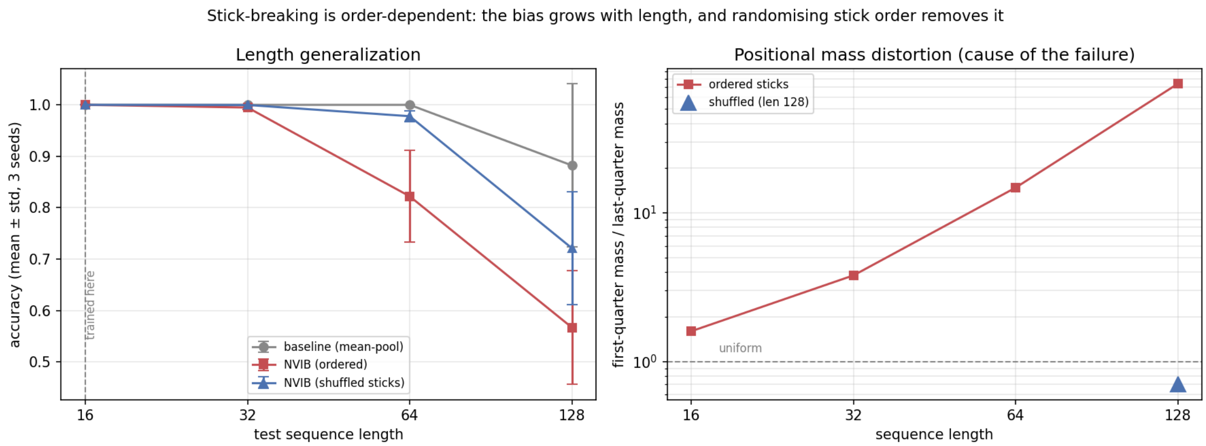

Figure 1: Left: length generalization for the baseline, the ordered bottleneck, and the causal control of Section 5. Right: the positional mass distortion that causes the ordered bottleneck's failure, log scale.

restoring exchangeability *in expectation* still leaves each individual forward pass with an arbitrary realized order, so some variance remains. The diagnosis is confirmed; the mechanism is improved, not fully repaired.

6 A Hyperparameter Sweep That Overturns the Interpretation

α_0 was held at 5 throughout the results above, under the assumption that it controlled the sparsity of the mixture — that raising it would concentrate weight further and worsen the positional distortion just measured. Sweeping $\alpha_0 \in \{1, 5, 20, 50\}$ produces the opposite of that prediction at every value tested (Table 2, Figure 2).

α_0	Accuracy @ 128	vs. baseline (0.916)	Position ratio @ 128	Top-1 hit @ 128
1	0.486 ± 0.066	-0.430	$3978\times$	0.129
5	0.556 ± 0.059	-0.360	$3343\times$	0.160
20	0.851 ± 0.078	-0.065	$23.1\times$	0.426
50	$\mathbf{0.944 \pm 0.053}$	$\mathbf{+0.028}$	$\mathbf{3.9}\times$	**0.862**

Table 2: Accuracy at $8\times$ training length rises monotonically with α_0; at $\alpha_0 = 50$ the ordered mechanism, with no shuffling at all, slightly exceeds the mean-pooling baseline.

The prior on each stick proportion is $\mathrm{Beta}(1, \alpha_0)$, with $\mathbb{E}[v] = 1/(1+\alpha_0)$: a larger α_0 makes each break take a *smaller* expected slice, so the stick survives more breaks and mass spreads across *more* positions before it is exhausted. Larger α_0 therefore makes the mixture more uniform, not sparser — the opposite of the assumption the initial experiment was run under, and consistent with how a concentration parameter behaves in the broader family of constructions this prior belongs to (a larger concentration parameter yields more, not fewer, effectively-used components).

7 Discussion

The corrected reading is not "the mechanism fails," nor simply "a hyperparameter was mistuned." It is that this prior's concentration sets an effective reach — informally, $\sim (1+\alpha_0)$ positions worth of stick-breaking budget — and the mechanism generalizes to sequences longer than training only when that reach exceeds the test length. The α_0 sweep is a dose-response curve for exactly that

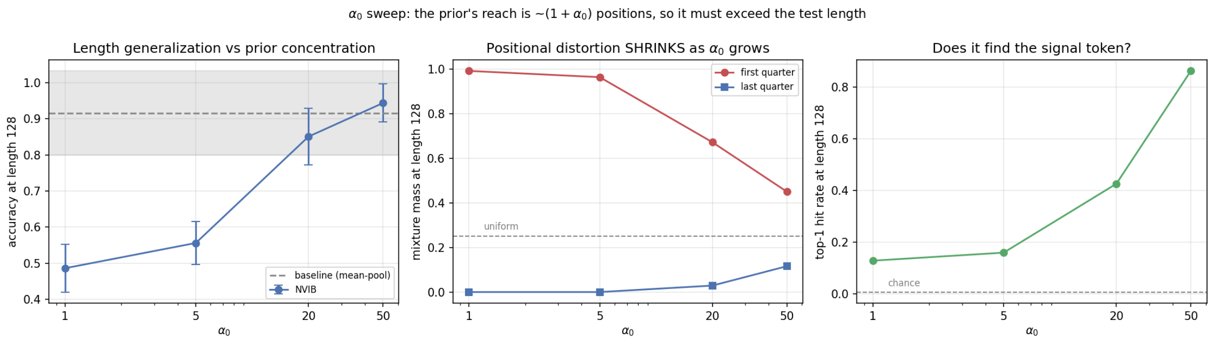

Figure 2: The α_0 sweep. Left: accuracy recovers to baseline parity by $\alpha_0 = 50$. Center: the positional distortion diagnosed in Section 4 shrinks monotonically. Right: the model's ability to identify the true signal token *improves* alongside it.

quantity: reach of roughly 2, 6, 21, and 51 positions yields accuracy 0.486, 0.556, 0.851, and 0.944 at a fixed test length of 128.

This also motivates why an adaptive concentration — one that scales with the input rather than being fixed by hand — would remove the need to retune α_0 per sequence length at all; a fixed value is a proxy for that adaptivity, and this experiment quantifies what a poorly-chosen fixed value costs.

8 Limitations

- One synthetic task, one architecture, one KL weight (0.01, never tuned). The concentration numbers in Section 4 confirm the bottleneck is binding rather than the KL term being negligible, but the specific weight is arbitrary.
- The α_0 sweep evaluates a single training length (16); the claim that reach must exceed test length is inferred from one training length, not measured across several.
- The shuffle control restores exchangeability in expectation only; each individual forward pass still imposes an arbitrary realized order.
- Every number here concerns this specific stick-breaking construction on this synthetic task at this scale. Generalization beyond that requires further testing, not assumption.

9 Path Forward

Two extensions follow directly from Section 7's reframing. First, sweeping α_0 jointly with test length and checking whether accuracy collapses onto a single curve when plotted against α_0/length would convert "reach must exceed length" from a qualitative reading into a tested scaling law. Second, making α_0 an output of the encoder rather than a fixed constant is a small architectural change and is the practical version of the input-adaptive concentration that Section 7 describes only as a design principle. Both are direct, low-cost follow-ons to the experiments reported here.

10 Reproducibility

Code, all four α_0 configurations, the shuffle-control run, and the figure-generating scripts: `https://github.com/saqarkpr/stickbreaking-attention-bottleneck`.